

Dark Energy as the Thermodynamic Afterglow of a Hidden Gauge-Theory Transition

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Abstract

DESI DR2 and Pantheon+ sharpen the case that late-time cosmology may require a dynamic dark sector, yet in standard General Relativity such dynamics are often modeled by phantom crossings, finely tuned masses, or ad hoc interaction shells. We study a different starting point. If gravity is a macroscopic thermodynamic equation of state in Jacobson's sense, then a Lorentz-invariant equilibrium vacuum contributes no null heat flow and must be removed from the macroscopic source law. Gravity is therefore sourced by the non-equilibrium stress tensor $\tau_{\mu\nu}$, not by the equilibrium baseline. Cosmic acceleration then requires a persistent out-of-equilibrium cosmic medium.

We model that source explicitly as the slowly relaxing thermodynamic exhaust of a hidden pure $SU(2)_D$ sector that confines at a dynamically generated meV scale. Below confinement the infrared medium is gapped and must be described causally by a memory variable Σ . The homogeneous arrested branch yields $\rho_X = 3c_D\Sigma$, $p_X = -(3c_D - 1)\Sigma$, and an intrinsic non-phantom equation of state $w_X = -1 + 1/(3c_D)$. Matter loading renormalizes the relaxation rate through a fixed signed kernel $\Psi(r) = 4r(r - 1)/(1 + r)^3$, producing the exchange law $Q = -(\beta/c_D)H\Psi(r)\rho_X$ and the effective equation of state $w_{\text{eff}} = -1 + [1 - \beta\Psi(r)]/(3c_D)$. The framework is causal, interacting, and explicitly testable in CLASS/MontePython, with an exact $\beta = 0$ branch, a mild effective phantom window, and a clear falsification path.

1. Introduction

The cosmological constant problem becomes clearer when it is split into two separate questions:

- (i) Why do the large equilibrium vacuum contributions of quantum fields and phase transitions fail to curve spacetime catastrophically?
- (ii) Why does the late universe nevertheless contain an accelerating component with present scale $\rho_{X,0}^{1/4} \simeq 2.3 \text{ meV}$?

Those questions need not have the same answer. In the framework studied here, they do not.

The standard presentation of the problem assumes from the start that gravity is fundamentally an action-principle geometry sourced by the full stress tensor. But if gravity is instead a macroscopic thermodynamic equation of state, the source law must be re-read. In Jacobson's horizon derivation, the relevant local quantity is the null heat flow across causal horizons, and a Lorentz-invariant equilibrium vacuum contributes exactly zero to that flow. On that reading, the vacuum catastrophe is not something to be hidden after the fact. It is a sign that the macroscopic source of gravity has been written incorrectly.

This changes the late-time dark-energy problem in a useful way. Once the equilibrium baseline is subtracted from the source term, the observed acceleration can no longer be attributed to an

exactly equilibrium vacuum. It must come from a medium that is still out of equilibrium today. Within that premise, a persistent cosmic non-equilibrium source is not optional. It is mandatory.

The model studied here supplies that source with three linked ingredients. First, the macroscopic source of gravity is the non-equilibrium stress tensor $\tau_{\mu\nu}$, not the equilibrium vacuum baseline. Second, a hidden pure $SU(2)_D$ sector generates a meV confinement scale dynamically through dimensional transmutation [2]. Third, after confinement the infrared dark sector is modeled as a gapped arrested medium whose long-wavelength state is characterized by a scalar pressure-deficit variable Σ . The exact-equilibrium remainder of that medium is subtracted into the equilibrium baseline, while the slowly relaxing non-equilibrium tail gravitates.

The isolated arrested phase already has a sharp consequence:

$$w_X = -1 + \frac{1}{3c_D}, \quad (1)$$

with c_D the late-time Deborah number. That is enough to produce an accelerating sourced branch when $c_D > 1/2$. The open-system extension developed here is designed to preserve that homogeneous branch while allowing the surrounding dark matter environment to modify the *rate* at which the tail relaxes and the *sign* of the exchange across the matter-dominated and dark-energy-dominated eras.

A recent multi-model comparison using DESI DR2, Pantheon+, Planck, and ACT provides a useful qualitative rubric for what late-time dark-energy theories should at least be able to mimic: a phantom tendency at higher redshift, a quintessence tendency at lower redshift, and possible late-time transfer from dark energy to matter, while neither decisively beating Λ CDM in Bayesian evidence nor resolving the Hubble tension outright [14]. We use that pattern only as a phenomenological target. This paper does *not* present a fit to those data.

What is exact and what is assumed should therefore be separated carefully.

- Exact within horizon thermodynamics: an equilibrium Lorentz-invariant vacuum drops out of the null source and is removed by equilibrium-subtraction from the macroscopic equation of state.
- Exact on the homogeneous post-transition arrested branch once $\tau_\Sigma = c_D/H$ is assumed: the confined phase splits as $\rho_{\text{conf}} = V_{D,\text{eq}} + 3c_D\Sigma$ and the sourced branch obeys (1).
- Constitutive but sharply restricted: the open-system correction depends on one dimensionless amplitude β and one fixed sign-changing kernel $\Psi(r)$.
- Constitutive and to be tested numerically: the late-time scaling $\tau_\Sigma = c_D/H$, the EFT activation of the post-confinement phase, the perturbative transfer frame, and the baseline scalar closure coefficient encoded in the bulk-mode speed c_b^2 .

The point of the paper is therefore limited but concrete. We do not claim a completed data fit. We do claim that the theory can be written in a form that is explicit enough to code, test, and falsify. The model uses one intrinsic constitutive parameter c_D , one interaction amplitude β , one fixed signed kernel $\Psi(r)$, and one independent scalar perturbation variable $\delta\Sigma$. It does not require a free response shell with multiple adjustable functions.

The paper is organized as follows. Section 2 redefines the source of gravity in thermodynamic language. Section 3 introduces the Yang-Mills trigger that makes a persistent non-equilibrium source plausible. Section 4 applies causal thermodynamics to the post-confinement medium and derives the interacting afterglow. Section 5 formulates the cosmological equations, perturbations, and CLASS/MontePython implementation strategy. Section 6 concludes.

2. The source of gravity

2.1 Horizon thermodynamics and equilibrium-subtraction

This section is a condensed version of the argument developed in [4], building on Jacobson's thermodynamic derivation of gravity [3] and the broader emergent-gravity perspective of Padmanabhan [6, 7, 8].

At any spacetime point one may consider the local causal horizon associated with an accelerated observer. In Jacobson's argument, the relevant heat flow across that horizon is controlled by the null projection

$$T_{\mu\nu}k^\mu k^\nu, \quad (2)$$

where k^μ is tangent to the null generators. A Lorentz-invariant equilibrium vacuum has the form

$$T_{\mu\nu}^{\text{eq}} = -\rho_{\text{eq}}g_{\mu\nu}. \quad (3)$$

Contracting with any null vector immediately gives

$$T_{\mu\nu}^{\text{eq}}k^\mu k^\nu = -\rho_{\text{eq}}g_{\mu\nu}k^\mu k^\nu = 0. \quad (4)$$

The equilibrium vacuum is therefore invisible to the null heat flow that underlies the macroscopic equation of state.

This suggests the natural macroscopic split

$$T_{\mu\nu} = T_{\mu\nu}^{\text{eq}} + \tau_{\mu\nu}, \quad \tau_{\mu\nu} := T_{\mu\nu} - T_{\mu\nu}^{\text{eq}}, \quad (5)$$

so that the local thermodynamic source depends only on the non-equilibrium stress tensor $\tau_{\mu\nu}$. The corrected macroscopic field equation is then

$$G_{\mu\nu} + \Lambda_{\text{ref}}g_{\mu\nu} = \frac{8\pi G}{c^4} \tau_{\mu\nu}. \quad (6)$$

The constant Λ_{ref} labels the curvature of the equilibrium branch. In this paper we work on the flat branch,

$$\Lambda_{\text{ref}} = 0, \quad G_{\mu\nu} = \frac{8\pi G}{c^4} \tau_{\mu\nu}. \quad (7)$$

This is not merely an algebraic reshuffling of a cosmological constant. The physical claim is that the equilibrium baseline is removed *before* the macroscopic source law is written.

Only a separately conserved equilibrium sector is subtracted in this way. Dynamical fields that are rolling, dissipating, or exchanging energy remain part of $\tau_{\mu\nu}$. In particular, inflation driven by a

rolling scalar is not erased by Eq. (6); only its strictly constant equilibrium baseline would drop out, exactly as in the argument of [4].

2.2 The mandate for a persistent non-equilibrium source

Equation (7) reorganizes the late-time problem in a sharp way. If the observed acceleration is not sourced by the equilibrium vacuum, then the universe must contain some medium whose non-equilibrium stress remains cosmologically relevant today. A successful dark-energy model in this framework cannot be a clever way of hiding the vacuum. It must be a concrete mechanism for sustaining a long-lived $\tau_{\mu\nu}$.

That observation sets the target for the rest of the paper. The source must satisfy three conditions. It must be near vacuum so that it can drive acceleration. It must be genuinely non-equilibrium so that it belongs on the right-hand side of Eq. (7). And it must relax slowly enough to survive for billions of years. Ordinary Standard Model plasmas fail the last requirement. We therefore look to a hidden confining sector whose post-transition medium is gapped and thermodynamically arrested.

3. Sourcing the friction: a hidden Yang-Mills sector

3.1 Why the source cannot be ordinary matter

The mandate from Section 2 is already restrictive. The source of today's acceleration must be a cosmic medium that is persistently out of equilibrium, yet close enough to vacuum to drive accelerated expansion. Ordinary visible-sector plasmas are the wrong objects: they thermalize too quickly, their late-time equation of state is not vacuum-like, and they do not naturally carry a billion-year relaxation time. What is needed instead is a medium that experienced a large thermodynamic shock, then entered a phase with a gapped spectrum and suppressed equilibration.

A confining hidden gauge theory supplies precisely that logic. Above confinement it behaves as a relativistic dark-radiation bath. Near confinement it undergoes a strongly non-conformal transition. Below confinement its infrared degrees of freedom are gapped, so the medium can retain memory of the transition and relax only slowly. In that setting, a persistent dark-sector contribution to $\tau_{\mu\nu}$ is not a guess added for convenience. It is the most economical way to satisfy the mandate created by Eq. (7).

3.2 Microscopic dark sector

The hidden sector is taken to be pure Yang-Mills with no light matter,

$$\mathcal{L}_D = -\frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu}, \quad a = 1, 2, 3. \quad (8)$$

Its one-loop beta function is

$$\mu \frac{dg_D}{d\mu} = -\frac{11N}{48\pi^2} g_D^3. \quad (9)$$

For pure $SU(2)_D$,

$$b_0 = \frac{11N}{3} = \frac{22}{3}, \quad (10)$$

and integrating the renormalization-group equation gives

$$\Lambda_D = \mu \exp \left[-\frac{8\pi^2}{b_0 g_D^2(\mu)} \right] = \mu \exp \left[-\frac{2\pi}{b_0 \alpha_D(\mu)} \right]. \quad (11)$$

With $\mu = M_{\text{Pl}} = 2.435 \times 10^{18} \text{ GeV}$ and $\Lambda_D \sim 2.3 \text{ meV}$ one finds

$$\alpha_D(M_{\text{Pl}}) \simeq \frac{2\pi}{b_0 \ln(M_{\text{Pl}}/\Lambda_D)} \simeq 0.0124. \quad (12)$$

Thus the required meV scale can arise dynamically through dimensional transmutation rather than by fine-tuning a rigid microscopic mass parameter.

3.3 Pre-confinement thermal history

Let the dark and visible sectors have temperature ratio

$$\xi \equiv \frac{T_D}{T_\gamma}. \quad (13)$$

Above confinement, the hidden sector is a bath of dark gluons with

$$g_D = 2(N^2 - 1) = 6 \quad (14)$$

relativistic bosonic degrees of freedom. Its contribution to the effective number of relativistic species is

$$\Delta N_{\text{eff}} = \frac{4}{7} g_D \left(\frac{T_D}{T_\nu} \right)^4 = \frac{4}{7} g_D \left(\frac{11}{4} \right)^{4/3} \xi^4 = 13.2 \xi^4. \quad (15)$$

The confinement transition occurs when

$$T_{D,c} = x_c \Lambda_D, \quad x_c = \mathcal{O}(1), \quad (16)$$

so the visible-sector redshift of confinement is

$$1 + z_c = \frac{T_{\gamma,c}}{T_{\gamma,0}} = \frac{x_c \Lambda_D}{\xi T_{\gamma,0}} \simeq 10.2 x_c \left(\frac{\Lambda_D}{2.3 \text{ meV}} \right) \xi^{-1}. \quad (17)$$

Equations (15) and (17) form the basic bridge between the microscopic hidden sector and the late-time effective description.

4. The arrested medium

4.1 Memory variable, equilibrium matching, and isolated branch

Below confinement we model the coarse-grained confined phase by an isotropic stress tensor

$$T_{\text{conf}}^{\mu\nu} = \rho_{\text{conf}} u^\mu u^\nu + p_{\text{conf}} \Delta^{\mu\nu}, \quad \Delta^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu, \quad (18)$$

with constitutive relation

$$p_{\text{conf}} = -\rho_{\text{conf}} + \Sigma. \quad (19)$$

The scalar $\Sigma \geq 0$ measures the pressure deficit away from exact equilibrium. Here it is not introduced as a convenient parametrization. It is the causal memory variable required by Israel-Stewart-type thermodynamics once the post-confinement medium is gapped and cannot equilibrate instantaneously [9, 10, 11].

In the isolated arrested regime the order parameter free-relaxes with finite relaxation time,

$$u^\mu \nabla_\mu \Sigma = -\frac{\Sigma}{\tau_\Sigma} = -\frac{\Theta}{3c_D} \Sigma, \quad \Theta \equiv \nabla_\mu u^\mu, \quad (20)$$

where the minimal late-time constitutive choice is

$$\tau_\Sigma = \frac{c_D}{H}, \quad c_D > 0. \quad (21)$$

In homogeneous FRW, where $\Theta = 3H$, Eq. (20) becomes

$$\dot{\Sigma} = -\frac{H}{c_D} \Sigma. \quad (22)$$

If the confined phase is isolated after the transition, local conservation gives

$$\dot{\rho}_{\text{conf}} + 3H(\rho_{\text{conf}} + p_{\text{conf}}) = 0 \quad \implies \quad \dot{\rho}_{\text{conf}} = -3H\Sigma. \quad (23)$$

Combining (22) and (23) yields

$$\frac{d\rho_{\text{conf}}}{d\Sigma} = 3c_D, \quad (24)$$

and therefore

$$\rho_{\text{conf}} = \rho_* + 3c_D \Sigma. \quad (25)$$

The integration constant is not left uninterpreted. In exact equilibrium $\Sigma \rightarrow 0$, Eq. (19) gives $p_{\text{conf}} \rightarrow -\rho_{\text{conf}}$, while Eq. (23) implies $\dot{\rho}_{\text{conf}} \rightarrow 0$. The residual piece is therefore vacuum-like,

$$T_{\text{eq}}^{\mu\nu} = -V_{D,\text{eq}} g^{\mu\nu}, \quad \delta V_{D,\text{eq}} = 0, \quad (26)$$

and we identify

$$\rho_* \equiv V_{D,\text{eq}}. \quad (27)$$

The sourced non-equilibrium tail is then defined by the equilibrium-subtracted stress tensor

$$\tau_X^{\mu\nu} := T_{\text{conf}}^{\mu\nu} - T_{\text{eq}}^{\mu\nu}. \quad (28)$$

The homogeneous post-transition branch is therefore

$$\rho_{\text{conf}} = V_{D,\text{eq}} + 3c_D \Sigma, \quad (29)$$

with sourced variables

$$\rho_X \equiv \rho_{\text{conf}} - V_{D,\text{eq}} = 3c_D \Sigma, \quad p_X \equiv p_{\text{conf}} + V_{D,\text{eq}} = -(3c_D - 1)\Sigma. \quad (30)$$

Its intrinsic equation of state is

$$w_X \equiv \frac{p_X}{\rho_X} = -1 + \frac{1}{3c_D}. \quad (31)$$

Thus the sourced tail is automatically non-phantom for every $c_D > 0$, and it accelerates whenever $c_D > 1/2$.

4.2 Derivation of the signed loading-unloading kernel

The post-confinement medium is not relaxing in a void. It shares the universe with cold dark matter, so its relaxation rate must feel an environmental load. At the homogeneous level the only natural low-derivative scalar for that load is the density ratio

$$r \equiv \frac{\rho_c}{\rho_X}. \quad (32)$$

A sign-changing kernel should satisfy five requirements.

1. It should be dimensionless and bounded.
2. It should vanish when either component becomes negligible, $r \rightarrow 0$ or $r \rightarrow \infty$.
3. It should change sign when the system passes through dark-sector balance.
4. It should be positive in the matter-loaded regime $r > 1$ and negative in the dark-energy-dominated regime $r < 1$.
5. It should be the minimal smooth polynomial choice once the above conditions are imposed.

Write

$$x \equiv \frac{r}{1+r} \in [0, 1]. \quad (33)$$

Then $x = 0$ and $x = 1$ correspond to $r = 0$ and $r = \infty$, while $x = 1/2$ corresponds to $r = 1$. The minimal polynomial that vanishes at all three points and is odd about $x = 1/2$ is

$$\Psi(x) = 4x(1-x)(2x-1). \quad (34)$$

Rewriting in terms of r gives the fixed signed kernel

$$\Psi(r) = \frac{4r(r-1)}{(1+r)^3}. \quad (35)$$

It obeys

$$\Psi(r) > 0 \quad (r > 1), \quad \Psi(r) < 0 \quad (0 < r < 1), \quad \Psi(1) = 0, \quad (36)$$

and vanishes in both asymptotic regimes. The kernel is fixed once the symmetry and sign requirements are chosen; it is not an independently sampled function.

4.3 Mandatory open-system loading and derived exchange law

The open-system postulate of the theory is that matter loading renormalizes the relaxation rate of the same memory variable Σ :

$$u^\mu \nabla_\mu \Sigma = -\frac{\Theta}{3c_D} [1 - \beta \Psi(r)] \Sigma, \quad \beta \geq 0. \quad (37)$$

In homogeneous FRW this becomes

$$\dot{\Sigma} = -\frac{H}{c_D} [1 - \beta \Psi(r)] \Sigma. \quad (38)$$

The parameter β measures the environmental loading-unloading correction relative to the isolated relaxation rate. The branch $\beta = 0$ returns exactly to the isolated sourced tail.

Now treat the sourced tail as an open subsystem,

$$\nabla_\mu \tau_X^{\mu\nu} = -Q^\nu, \quad \nabla_\mu T_c^{\mu\nu} = Q^\nu. \quad (39)$$

In homogeneous FRW only the scalar Q enters, so

$$\dot{\rho}_X + 3H(\rho_X + p_X) = -Q \quad \implies \quad \dot{\rho}_X + \frac{H}{c_D} \rho_X = -Q. \quad (40)$$

Combining (38), (40), and the requirement that the *homogeneous* branch relation (29) continue to hold in the open system gives

$$\frac{d\rho_{\text{conf}}}{d\Sigma} = \frac{-3H\Sigma - Q}{-\frac{H}{c_D}(1 - \beta\Psi)\Sigma} = 3c_D. \quad (41)$$

Solving (41) for Q yields the background exchange law

$$Q = -3\beta H \Psi(r) \Sigma = -\frac{\beta}{c_D} H \Psi(r) \rho_X. \quad (42)$$

This is the key structural point: once the homogeneous branch relation is preserved, the exchange law is not an independent ansatz. It is determined by the loading-unloading relaxation law.

The background evolution equations are therefore

$$\dot{\rho}_X = -\frac{H}{c_D} \left[1 - \beta \Psi(r) \right] \rho_X, \quad (43)$$

$$\dot{\rho}_c + 3H\rho_c = -\frac{\beta}{c_D} H \Psi(r) \rho_X. \quad (44)$$

Hence the sourced relation $\rho_X = 3c_D \Sigma$ remains exact at the homogeneous background level in the interacting arrested regime.

4.4 Intrinsic and effective equations of state, and the sign of transfer

The physical stress tensor of the sourced tail is fixed by Eq. (31). Observers, however, often fit background data with an uncoupled dark-energy fluid defined by

$$\dot{\rho}_X + 3H(1 + w_{\text{eff}}) \rho_X = 0. \quad (45)$$

Comparing (43) and (45) gives

$$w_{\text{eff}} = -1 + \frac{1 - \beta \Psi(r)}{3c_D}. \quad (46)$$

The intrinsic and effective equations of state therefore differ by

$$w_{\text{eff}} - w_X = -\frac{\beta}{3c_D} \Psi(r). \quad (47)$$

Three immediate consequences follow.

1. In the matter-loaded regime $r > 1$, one has $\Psi(r) > 0$, so $w_{\text{eff}} < w_X$. If $\beta \Psi(r) > 1$, then $w_{\text{eff}} < -1$ even though the physical branch remains non-phantom.
2. At dark-sector balance $r = 1$, one has $\Psi(1) = 0$, so $w_{\text{eff}} = w_X$ and $Q = 0$.
3. In the dark-energy-dominated regime $0 < r < 1$, one has $\Psi(r) < 0$, so $w_{\text{eff}} > w_X > -1$ and

$$Q > 0, \quad (48)$$

meaning that energy flows from dark energy to matter.

This is the qualitative sign pattern targeted by the model: effective phantom at higher redshift, quintessence-like behavior at lower redshift, and late-time dark-energy-to-matter transfer, all with a non-phantom intrinsic branch.

It is also useful to record the evolution equation for r in $N \equiv \ln a$:

$$\frac{dr}{dN} = r \left(-3 + \frac{1}{c_D} \right) - \frac{\beta}{c_D} \Psi(r) (1 + r). \quad (49)$$

This equation makes the sign structure transparent. For $r > 1$, the interaction term pushes r downward faster. For $r < 1$, it pushes r upward. The late-time branch is therefore not automatic; its stability must be checked.

Because the kernel has a bounded maximum,

$$\Psi_{\max} = \frac{2\sqrt{3}}{9}, \quad (50)$$

one also has

$$w_{\text{eff}}^{\min} = -1 + \frac{1 - \beta\Psi_{\max}}{3c_D}. \quad (51)$$

Within the baseline crossing strip discussed in Section 5 and for the near-vacuum values $c_D \gtrsim 5$ favored for a late arrested phase, the effective phantom excursion is therefore expected to be mild, typically at the level of a few times 10^{-3} to 10^{-2} below -1 , not a deep phantom regime.

4.5 Asymptotics and the non-phantom future

Because the kernel vanishes in both asymptotic regimes,

$$\Psi(r) \rightarrow 0 \quad \text{for} \quad r \rightarrow \infty \quad \text{or} \quad r \rightarrow 0, \quad (52)$$

the isolated tail is recovered both early and late. The far-future branch relevant to this paper is the dark-energy-dominated one, $r \rightarrow 0$. Expanding Eq. (49) for small r gives

$$\Psi(r) = -4r + \mathcal{O}(r^2), \quad \frac{dr}{dN} = r \left[-3 + \frac{1 + 4\beta}{c_D} \right] + \mathcal{O}(r^2). \quad (53)$$

Therefore the decaying late-time branch $r \rightarrow 0$ is an attractor provided

$$c_D > \frac{1 + 4\beta}{3}. \quad (54)$$

This condition is mild. It is automatically satisfied throughout the near-vacuum regime $c_D \gtrsim 5$ even on the signed-kernel crossing branch.

Under Eq. (54), the late-time asymptotics reduce to the isolated tail,

$$\rho_X \propto a^{-1/c_D}, \quad w_{\text{eff}} \rightarrow w_X = -1 + \frac{1}{3c_D}. \quad (55)$$

If the sourced tail dominates the late universe, the Friedmann equation gives

$$a(t) \propto t^{2c_D}, \quad \rho_X(t) \propto t^{-2}. \quad (56)$$

The late future is therefore a slowly relaxing power-law acceleration sourced by the non-equilibrium tail itself, not a de Sitter state produced by an equilibrium vacuum. The model allows a transient effective $w_{\text{eff}} < -1$ episode, but not a true asymptotically phantom attractor.

5. Cosmological implementation and falsification path

The equations in this section track the evolution of the non-equilibrium stress tensor $\tau_{\mu\nu}$ as it interacts with the cold-dark-matter environment. At background level this means ρ_X and p_X are

the dark-sector contribution to $\tau_{\mu\nu}$. At perturbative level it means the memory variable must be evolved explicitly rather than collapsed into an instantaneous fluid closure.

5.1 Background FRW equations and transition activation

On the flat branch of Eq. (7), the local cosmological equations take the standard FRW form,

$$3M_{\text{Pl}}^2 H^2 = \rho_b + \rho_c + \rho_r + \rho_X + \rho_{dr,D}, \quad (57)$$

$$-2M_{\text{Pl}}^2 \dot{H} = \rho_b + \rho_c + \frac{4}{3}\rho_r + \frac{4}{3}\rho_{dr,D} + \rho_X + p_X, \quad (58)$$

where each term on the right-hand side belongs to the active non-equilibrium source and $\rho_{dr,D}$ denotes the pre-confinement dark-radiation component. The sourced tail satisfies

$$\rho_X + p_X = \Sigma = \frac{\rho_X}{3c_D}. \quad (59)$$

The novel background equations are therefore Eqs. (43), (44), and (35).

The arrested-phase effective theory begins after confinement. What we model here is the activation of the *post-confinement effective theory*, not a first-principles microphysical description of the confinement transition itself. It is therefore helpful to separate the bookkeeping from the constitutive late-time dynamics.

Let the pre-confinement dark-radiation energy density immediately before confinement be $\rho_{dr,D}(a_c^-)$. A general transition bookkeeping may be written as

$$\rho_{dr,D}(a_c^-) = \rho_{X,c}^{\text{init}} + V_{D,\text{eq}}^{\text{init}} + \rho_{m,D}^{\text{init}}, \quad (60)$$

with fractional decomposition

$$f_X + f_{\text{eq}} + f_m = 1, \quad \rho_{X,c}^{\text{init}} = f_X \rho_{dr,D}(a_c^-), \quad V_{D,\text{eq}}^{\text{init}} = f_{\text{eq}} \rho_{dr,D}(a_c^-). \quad (61)$$

Here f_X feeds the sourced arrested tail, f_{eq} feeds the exact-equilibrium remainder that returns to the equilibrium baseline, and f_m would describe any long-lived massive hidden relic component. The minimal version studied in this draft sets

$$f_m = 0, \quad (62)$$

and uses a single post-transition amplitude to encode the sourced branch,

$$\Sigma(a_c^+) = \varepsilon_\Sigma \Lambda_D^4, \quad \rho_X(a_c^+) = 3c_D \varepsilon_\Sigma \Lambda_D^4. \quad (63)$$

The exact-equilibrium remainder is then understood to return to $T_{\mu\nu}^{\text{eq}}$, not to survive as a separately gravitating FRW component.

If $\beta = 0$, the source-free branch is

$$\rho_X(a) = 3c_D \varepsilon_\Sigma \Lambda_D^4 \left(\frac{a}{a_c} \right)^{-1/c_D}. \quad (64)$$

For production Boltzmann runs it is usually safer to replace a sharp activation by the smooth profile

$$S(a) = \frac{1}{2} \left[1 + \tanh \left(\frac{\ln a - \ln a_c}{\Delta_c} \right) \right], \quad \Delta_c \ll 1. \quad (65)$$

Using $N \equiv \ln a$, a practical background implementation is

$$\frac{d\rho_c}{dN} = -3\rho_c - S(a) \frac{\beta}{c_D} \Psi(r) \rho_X, \quad (66)$$

$$\frac{d\Sigma}{dN} = -\frac{1 - \beta S(a) \Psi(r)}{c_D} \Sigma + \Sigma_c \frac{dS}{dN}, \quad \Sigma_c \equiv \varepsilon_\Sigma \Lambda_D^4, \quad (67)$$

$$\rho_X = 3c_D \Sigma, \quad r = \frac{\rho_c}{\rho_X}. \quad (68)$$

If the pre-confinement dark-radiation bath is to be switched off explicitly in the sourced subsystem, one may also use

$$\rho_{dr,D}(a) \rightarrow [1 - S(a)] \rho_{dr,D}(a). \quad (69)$$

Equations (66)-(69) are an EFT activation prescription for the post-confinement tail. They are not claimed to constitute a microphysical model of the phase transition.

5.2 Microscopic reconstruction and background-only benchmarks

The phenomenological late-time parameters map back to the microscopic picture through a short chain of relations. From Eq. (15),

$$\xi = \left(\frac{\Delta N_{\text{eff}}}{13.2} \right)^{1/4}. \quad (70)$$

Then Eq. (17) gives

$$\Lambda_D = \frac{\xi T_{\gamma,0}}{x_c} (1 + z_c). \quad (71)$$

The ultraviolet gauge coupling is

$$\alpha_D(M_{\text{Pl}}) = \frac{2\pi}{b_0 \ln(M_{\text{Pl}}/\Lambda_D)}, \quad b_0 = \frac{22}{3}. \quad (72)$$

Finally,

$$\varepsilon_\Sigma = \frac{\rho_X(a_c^+)}{3c_D \Lambda_D^4}. \quad (73)$$

In the source-free case,

$$\rho_X(a_c^+) = \rho_{X,0} (1 + z_c)^{1/c_D}. \quad (74)$$

In the signed-kernel case,

$$\rho_X(a_c^+) = \rho_{X,0} \exp \left[\int_{a_c}^1 \frac{1 - \beta \Psi[r(a)]}{c_D} d \ln a \right]. \quad (75)$$

This integral should be evaluated numerically on the sampled background. The microscopic interpretation only survives if the preferred phenomenological region maps back to sensible values

of $(\alpha_D(M_{\text{Pl}}), \xi, x_c, \varepsilon_\Sigma)$.

Before any CLASS or MCMC analysis, the background system can already be made concrete. We therefore present two illustrative benchmark backgrounds obtained by backward-integrating Eqs. (43) and (44) from $a = 1$ to $a_c = (1 + z_c)^{-1}$, using

$$(\Omega_{b,0}, \Omega_{c,0}, \Omega_{r,0}, \Omega_{X,0}) = (0.049, 0.261, 9.2 \times 10^{-5}, 0.689908), \quad (76)$$

flatness at $a = 1$, $x_c = 1$, and the common transition inputs $z_c = 200$ and $\Delta N_{\text{eff}} = 10^{-4}$. These are *illustrative* backgrounds, not a fit to data.

Benchmark	z_c	ΔN_{eff}	c_D	β	w_X	$w_{\text{eff}}^{\text{min}}$	ξ	Λ_D [meV]	$\alpha_D(M_{\text{Pl}})$	ε_Σ
A	200	10^{-4}	8.0	1.5	-0.9583	-0.9824	0.0525	2.476	0.0124	0.0519
B	200	10^{-4}	5.5	3.0	-0.9394	-1.0094	0.0525	2.476	0.0124	0.0937

Table 1: Illustrative background benchmark points obtained by backward integration from $a = 1$. Benchmark A is the non-crossing branch. Benchmark B crosses below $w_{\text{eff}} = -1$ only mildly, between $z \approx 0.96$ and $z \approx 1.79$, while remaining intrinsically non-phantom.

Figure 1 shows the resulting background evolution. Benchmark A stays on the non-crossing branch. Benchmark B enters a mild effective phantom phase, but only at the level of $\sim 10^{-2}$ below -1 , exactly as anticipated from Eq. (51). The exchange term changes sign near $r = 1$, and the expansion rate differs from the corresponding Λ CDM benchmark by at most a few percent over the range shown.

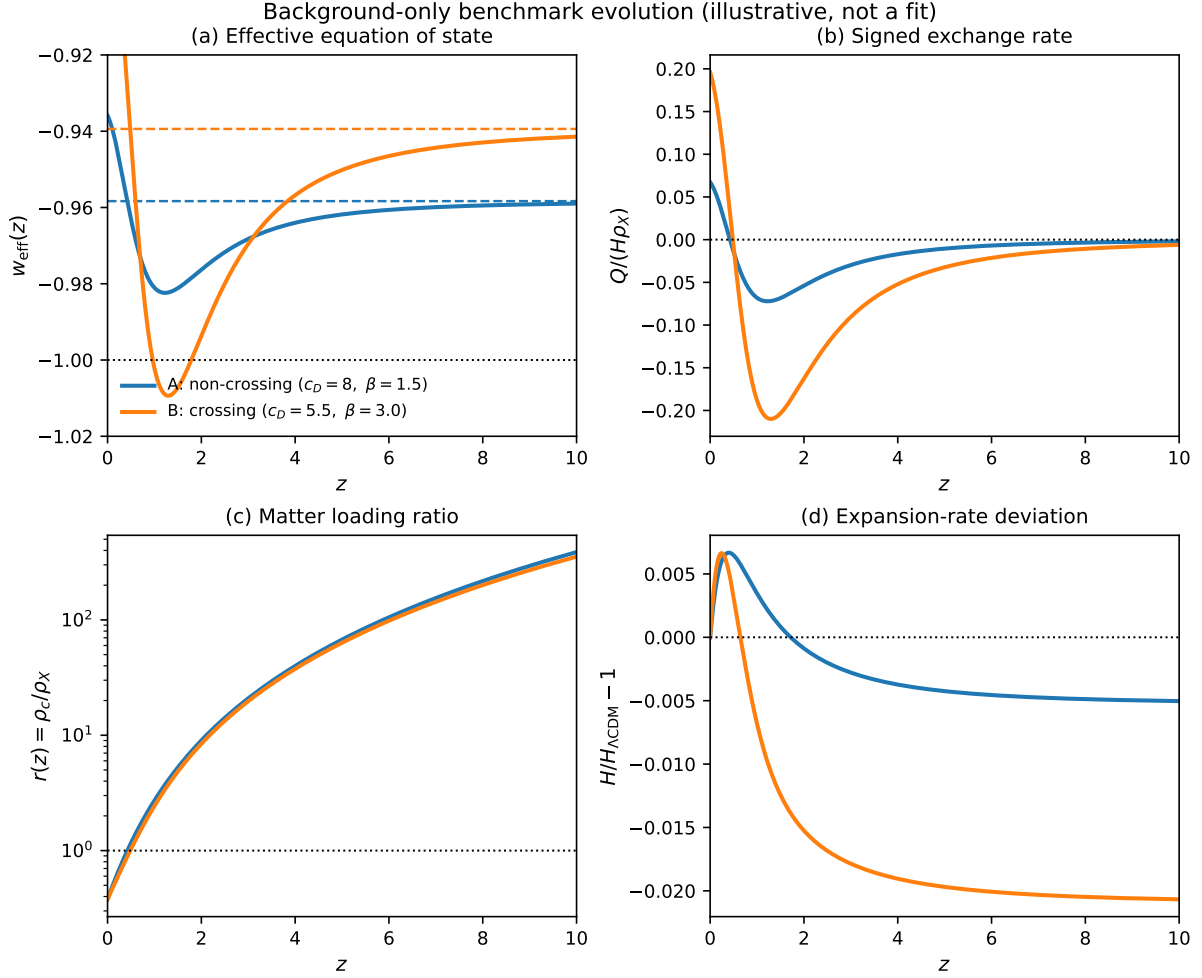


Figure 1: Background-only benchmark evolution for the two illustrative points in Table 1. The dashed horizontal lines in panel (a) show the intrinsic constant values w_X for the two benchmarks; the dotted line marks $w = -1$. Panel (b) shows the sign-changing exchange rate, panel (c) the loading ratio $r(z)$, and panel (d) the deviation of the Hubble rate from a flat Λ CDM model with the same present-day $(\Omega_{b,0}, \Omega_{c,0}, \Omega_{r,0}, \Omega_{X,0})$.

These plots do not replace a Boltzmann analysis, but they are the right pre-CLASS benchmark: they make the background structure tangible, they show the sign logic explicitly, and they reveal that the allowed effective phantom phase is expected to be shallow rather than dramatic.

5.3 Linear perturbations

Why the stock fld module is not the theory

The theory contains a genuine memory variable, Σ . It is therefore not equivalent to specifying a background equation of state $w(a)$ and passing that function to the stock CLASS generalized-fluid module. Doing so would collapse a causal relaxation system into an instantaneous barotropic closure. That is not the model defined above. A correct Boltzmann treatment must evolve the perturbation

of the constitutive variable explicitly.

Background-exact identity versus independent memory mode

The most important conceptual clarification is simple. The relation

$$\bar{\rho}_X = 3c_D \bar{\Sigma} \quad (77)$$

is an exact *homogeneous FRW identity* of the post-transition arrested branch. It is not a local algebraic constitutive law that forces perturbations to satisfy $\delta\rho_X = 3c_D\delta\Sigma$ at every spacetime point.

Accordingly, in linear theory the perturbation of the memory variable is independent. Define

$$s \equiv \delta\Sigma, \quad \chi \equiv \delta\rho_X - 3c_D s. \quad (78)$$

The quantity χ measures the non-adiabatic mismatch between the density perturbation and the perturbation of the memory variable. In this framework χ need not vanish. The correct perturbative question is therefore not whether $\delta\rho_X$ can be reconstructed algebraically from s , but how χ evolves under the constitutive closure.

Baseline frame and dynamical variables

The signed-kernel background law does not by itself fix the perturbative transfer frame. For the first explicit implementation we adopt the standard baseline choice

$$Q^\nu = Q u_c^\nu, \quad (79)$$

which closes the scalar sector directly in standard synchronous gauge. This is a baseline implementation choice, not an additional response shell. Once the baseline system is stable, a re-derivation in the Landau frame of the combined dark sector is a worthwhile robustness check.

We work in synchronous gauge,

$$ds^2 = a^2(\eta) [-d\eta^2 + (\delta_{ij} + h_{ij})dx^i dx^j], \quad (80)$$

and choose the usual cold-dark-matter comoving gauge so that $\theta_c = 0$. Because the baseline transfer vector is aligned with u_c^μ , that condition is preserved.

For the sourced tail define

$$\nu_X \equiv (\bar{\rho}_X + \bar{p}_X)\theta_X = \bar{\Sigma}\theta_X, \quad s \equiv \delta\Sigma, \quad (81)$$

so that the momentum variable remains well defined even if $\bar{\Sigma}$ becomes small. The pressure perturbation follows directly from the constitutive relation,

$$\delta p_X = -\delta\rho_X + s. \quad (82)$$

This is the crucial difference from a barotropic fluid: δp_X is not fixed by $\delta \rho_X$ alone.

Energy, momentum, and baseline hyperbolic closure

Linearizing $\nabla_\mu T_c^{\mu\nu} = Q^\nu$ and $\nabla_\mu \tau_X^{\mu\nu} = -Q^\nu$ in the baseline frame (79) gives

$$\delta \rho'_c + 3\mathcal{H}\delta \rho_c + \bar{\rho}_c \frac{h'}{2} = a \delta Q, \quad (83)$$

$$\delta \rho'_X + 3\mathcal{H}s + \nu_X + \bar{\Sigma} \frac{h'}{2} = -a \delta Q, \quad (84)$$

$$\nu'_X + 4\mathcal{H}\nu_X - k^2(-\delta \rho_X + s) = -a\bar{Q} \frac{\nu_X}{\bar{\Sigma}}. \quad (85)$$

Because $a\bar{Q} = -3\beta\mathcal{H}\Psi(r)\bar{\Sigma}$, the Euler equation becomes

$$\nu'_X + \left[4 - 3\beta\Psi(r)\right]\mathcal{H}\nu_X - k^2(-\delta \rho_X + s) = 0. \quad (86)$$

The constitutive perturbation must reduce to the background relaxation law in the homogeneous limit while remaining hyperbolic at short wavelength. We therefore adopt the baseline closure ansatz

$$s' = -\frac{\mathcal{H}}{c_D}s - \frac{a\delta Q}{3c_D} - (1 + c_b^2) \left(\nu_X + \bar{\Sigma} \frac{h'}{2} \right), \quad 0 < c_b^2 \leq 1. \quad (87)$$

Here c_b^2 is the short-wavelength bulk-mode speed squared. Equation (87) is not claimed to be unique. It is the minimal one-parameter linear closure we adopt for first perturbative tests because it preserves the homogeneous law, evolves the memory variable explicitly, and gives a hyperbolic short-wave limit. The independent variable $s = \delta \Sigma$ must be evolved explicitly. It is not to be reconstructed algebraically from $\delta \rho_X$.

Perturbed interaction and short-wave limit

For the signed-kernel exchange law (42), the perturbation follows by varying the same local variables that define it. In the cold-dark-matter rest frame,

$$a \delta Q = -\beta\Psi(r) \left(3\mathcal{H}s + \bar{\Sigma} \frac{h'}{2} \right) - 3\beta\mathcal{H}\bar{\Sigma} \Psi_{,r} \delta r, \quad (88)$$

with

$$\delta r = \frac{1}{\bar{\rho}_X} (\delta \rho_c - r \delta \rho_X), \quad \Psi_{,r} = \frac{4(-r^2 + 4r - 1)}{(1 + r)^4}. \quad (89)$$

The short-wavelength, no-expansion, no-interaction limit determines the coefficient in Eq. (87). Setting $\mathcal{H} = \bar{Q} = h' = 0$ gives

$$\delta \rho'_X + \nu_X = 0, \quad \nu'_X - k^2(-\delta \rho_X + s) = 0, \quad s' = -(1 + c_b^2)\nu_X. \quad (90)$$

For plane waves $\propto e^{-i\omega\eta}$ this yields

$$\omega^2 = c_b^2 k^2. \quad (91)$$

Thus Eq. (87) has the desired hyperbolic short-wave limit.

Baseline-frame crossing strip and practical guards

The explicit synchronous-gauge system immediately yields the most important practical bound. Equation (86) shows that the local momentum-friction coefficient of the sourced tail is

$$\Gamma_\nu = [4 - 3\beta\Psi(r)]\mathcal{H}. \quad (92)$$

Since $\Psi(r) < 0$ for $r < 1$, the dark-energy-dominated regime is automatically safe in the explicit baseline frame. Anti-damping is possible only in the matter-loaded regime $r > 1$, where $\Psi(r) > 0$.

Transient effective phantom crossing requires $\beta\Psi(r) > 1$. The signed kernel reaches its positive maximum at $r = 2 + \sqrt{3}$, with

$$\Psi_{\max} = \frac{2\sqrt{3}}{9}. \quad (93)$$

Therefore crossing is possible only if

$$\beta > \frac{1}{\Psi_{\max}} = \frac{3\sqrt{3}}{2} \approx 2.60. \quad (94)$$

To avoid local anti-damping in the same regime one needs $\beta\Psi(r) < 4/3$. Using Eq. (93), a sufficient global guard is

$$\beta < \frac{4}{3\Psi_{\max}} = 2\sqrt{3} \approx 3.46. \quad (95)$$

Hence the explicit baseline implementation supports a baseline-frame crossing strip,

$$\frac{3\sqrt{3}}{2} < \beta < 2\sqrt{3}. \quad (96)$$

The remaining stability guards are straightforward:

1. $c_D > 1/2$ if the intrinsic sourced branch is to accelerate;
2. $c_D > (1 + 4\beta)/3$ if the late-time decaying branch $r \rightarrow 0$ is to be stable;
3. $\bar{\Sigma} > 0$ and $\bar{\rho}_X = 3c_D\bar{\Sigma} > 0$ at all times;
4. $0 < c_b^2 \leq 1$ so the short-wave system is hyperbolic and subluminal;
5. $\beta \geq 0$.

Within these guards, the model can support an effective crossing branch without ever making the intrinsic dark-energy sector phantom. The benchmark of Section 5 shows that the resulting excursion is typically mild, which is the physically relevant expectation to carry into the first CLASS runs.

5.4 CLASS and MontePython roadmap

The observational stress test is straightforward in principle even if it is technically demanding. The exact $\beta = 0$ branch is the first target: it must survive a full CMB+BAO+SNIa Boltzmann treatment before the interacting branch can be taken seriously. Only after that should the signed kernel be confronted with production MCMC chains.

The background code should add a dedicated sourced-tail species. The minimal set of background quantities is

$$\{\Sigma, \rho_X, r, \Psi, Q\}$$

plus the optional pre-confinement dark-radiation component $\rho_{dr,D}$ and the switch $S(a)$. The cleanest implementation is to evolve Σ and derive $\rho_X = 3c_D\Sigma$ algebraically. This preserves the defining *background* identity exactly by construction.

There are two practical amplitude strategies.

1. **Backward integration for debugging.** Start at $a = 1$ with

$$\rho_{X,0} = 3M_{\text{Pl}}^2 H_0^2 - \rho_{b,0} - \rho_{c,0} - \rho_{r,0} - \rho_{dr,D,0}, \quad (97)$$

set $\Sigma_0 = \rho_{X,0}/(3c_D)$, and integrate backward to reconstruct $\rho_X(a_c^+)$ and ε_Σ .

2. **Forward integration for production.** Use Σ_c or ε_Σ as a shooting parameter and adjust it until flatness at $a = 1$ is satisfied. The source-free analytic estimate Eq. (74) is a good initial guess.

Backward integration is often the fastest way to verify that the background equations are coded correctly. Forward integration is more natural inside CLASS once the model is stable.

The perturbation code should add at least the variables

$$\{\delta\rho_X, \nu_X, s\}$$

and evolve the system (83), (84), (86), (87), and (88). The single most important rule is simple:

Do not reconstruct $s = \delta\Sigma$ algebraically from $\delta\rho_X$.

The independent perturbation of the constitutive variable is the whole point of the model.

In practice it is also useful to monitor the diagnostic combination

$$\chi \equiv \delta\rho_X - 3c_D s \quad (98)$$

during development runs. The model predicts that χ evolves dynamically; it is not a numerical error to find $\chi \neq 0$.

For development runs it is sensible to fix

$$c_b^2 = \frac{1}{2} \quad (99)$$

and release c_b^2 only after the spectra are stable. If the transition occurs early, adiabatic initial conditions for the pre-confinement dark-radiation bath are usually sufficient and the switch generates the tail dynamically. For sharp-switch debugging, a simple matched post-transition choice is

$$s(a_c^+) = \frac{1}{3c_D} \delta\rho_X(a_c^+), \quad \nu_X(a_c^+) = 0. \quad (100)$$

This is not claimed to be unique. It is a controlled first choice for code validation.

A practical microscopic basis is

$$\{\alpha_D(M_{\text{Pl}}), \xi, x_c, \varepsilon_\Sigma, c_D, c_b^2, \beta\}, \quad (101)$$

while a better first-stage sampling basis is

$$\{z_c, \Delta N_{\text{eff}}, c_D, c_b^2, \beta\} \quad (102)$$

together with the standard cosmological parameters and either a flatness condition or a shooting parameter for the source amplitude. The phenomenological basis is numerically better conditioned. The microscopic reconstruction can be done afterward using Section 5.

The signed-kernel interaction should not be sampled on the full positive real line in first production chains. The most stable practice is to sample separate branches:

1. a **non-crossing branch** with $0 \leq \beta \leq 2.4$;
2. a **crossing branch** with $2.65 \lesssim \beta \lesssim 3.40$.

These ranges stay respectively below and inside the explicit baseline strip (96). In addition, every sampled point should satisfy the late-branch guard (54).

Table 2 lists a conservative initial prior box for first production chains.

Parameter	Start range	Comment
$\log_{10}(1 + z_c)$	[2, 4.5]	Keeps the transition well after BBN and usually before the observable onset of dark-energy domination.
$\log_{10} \Delta N_{\text{eff}}$	[−6, −1]	Starts from negligible dark radiation and allows a modest hidden-radiation bath.
$\log_{10} c_D$	[0.7, 2]	Corresponds to $c_D \in [5, 100]$; broad enough for theory testing while automatically satisfying $c_D > (1 + 4\beta)/3$ over the recommended β ranges.
c_b^2	fixed at 1/2 first; then [0.1, 1]	Hold fixed until the new perturbation sector is numerically stable.
β	[0, 2.4] or [2.65, 3.40]	Run non-crossing and crossing branches separately. The crossing branch stays inside the baseline anti-damping guard.

Table 2: Recommended starting priors for first-stage chains.

The numerically sane sequence is:

1. validate the isolated source-free branch $\beta = 0$ in the background;
2. add the smooth transition and dark-radiation switch while keeping $\beta = 0$;
3. add the perturbation variables $\delta\rho_X$, ν_X , and s on the non-crossing branch;
4. once the scalar sector is stable, run the separate crossing branch inside (96);
5. only after the baseline implementation is reliable, consider re-deriving the source terms in the total-dark-sector Landau frame as a robustness test.

Do not turn on every structural ingredient at once. Diagnosing failures is much easier when the memory variable, the transition, and the signed loading-unloading correction are introduced one at a time.

Before any serious MontePython run, the modified code should pass the following checks.

1. Recover the analytic source-free background scaling $\rho_X \propto a^{-1/c_D}$ when $\beta = 0$.
2. Preserve the background identity $\rho_X = 3c_D\Sigma$ to machine precision.
3. Recover Λ CDM continuously in the limit $c_D \rightarrow \infty$, $\Delta N_{\text{eff}} \rightarrow 0$, and $\beta \rightarrow 0$.
4. Verify numerically that $Q < 0$ for $r > 1$, $Q = 0$ at $r = 1$, and $Q > 0$ for $r < 1$.
5. Verify that $Q \rightarrow 0$ and $w_{\text{eff}} \rightarrow w_X$ in both asymptotic regimes $r \rightarrow \infty$ and $r \rightarrow 0$.
6. Show no gauge singularity as w_{eff} crosses -1 on the separate crossing branch.
7. Keep $\bar{\Sigma} > 0$ and $\bar{\rho}_X > 0$ for every accepted sample.
8. Enforce the explicit baseline-frame guard $4 - 3\beta\Psi(r) > 0$ over the full time window covered by the integrator.
9. Monitor $\chi = \delta\rho_X - 3c_D s$ as an evolved perturbative quantity rather than incorrectly forcing it to zero.
10. Run the full CMB+BAO+SNIa likelihood first on the exact $\beta = 0$ branch, then on the interacting branches.
11. Show convergence of CMB and matter power spectra with respect to the switch width Δ_c and the k -sampling of the new species.

Once a phenomenologically viable posterior exists, reconstruct the microscopic parameters sample by sample:

1. compute ξ from Eq. (70);
2. choose or profile over $x_c = \mathcal{O}(1)$ and compute Λ_D from Eq. (71);
3. compute $\alpha_D(M_{\text{Pl}})$ from Eq. (72);

4. compute $\rho_X(a_c^+)$ from the sampled background using Eq. (75);
5. infer ε_Σ from Eq. (73).

A posterior region that fits the data but maps to implausible microscopic values is not a success for the hidden-sector interpretation. It is only a successful late-time effective fit.

6. Conclusion

The logic of the paper has two halves that now lock together cleanly.

1. In horizon thermodynamics, a Lorentz-invariant equilibrium vacuum contributes no null heat flow and therefore drops out of the macroscopic source law. Gravity responds to $\tau_{\mu\nu}$, not to the equilibrium baseline.
2. Because the universe is accelerating, there must exist a cosmic medium that remains persistently out of equilibrium and is still near vacuum today.
3. A hidden pure Yang-Mills sector provides the natural trigger: dimensional transmutation generates the meV scale, confinement supplies the thermodynamic shock, and the gapped post-transition phase supplies the slow relaxation.
4. The resulting afterglow is described by one memory variable Σ , one interaction amplitude β , one fixed sign-changing kernel $\Psi(r)$, and one independent scalar perturbation variable $\delta\Sigma$.

This reformulation is more economical than a generic interacting-fluid shell. The sourced tail itself is never phantom. What can cross -1 is only the *effective* uncoupled fit, and even then only mildly inside the baseline crossing strip. After dark-energy domination, the sign of Ψ reverses, Q becomes positive, and the effective equation of state becomes more quintessence-like than the intrinsic branch. Because the kernel dies away automatically for $r \rightarrow 0$ and $r \rightarrow \infty$, the far future is always the isolated decaying tail.

The framework is also fully exposed to failure, which is exactly as it should be. If the perturbations are unstable, the model fails. If the $\beta = 0$ branch cannot survive full CMB+BAO+SN Ia chains, the basic arrested-medium idea fails. If stable posteriors map back to implausible microscopic parameters, the hidden-sector interpretation fails. And if none of those things happens, then the proposal becomes interesting for a concrete reason: within the thermodynamic reading of gravity, it supplies the mandatory non-equilibrium source with a compact and testable late-time effective theory.

The ball is now with the data and the MCMC solvers.

A. Useful exact background identities

From Eq. (30),

$$\rho_X = 3c_D \Sigma, \quad p_X = -(3c_D - 1)\Sigma, \quad \rho_X + p_X = \Sigma.$$

Together with Eq. (42) this gives

$$\dot{\rho}_X + \frac{H}{c_D} \rho_X + Q = 0, \quad \dot{\Sigma} + \frac{H}{c_D} \Sigma + \frac{Q}{3c_D} = 0. \quad (103)$$

These identities are exact within the post-transition signed-kernel effective theory at the homogeneous level.

B. Extrema of the signed kernel

Differentiating Eq. (35) gives

$$\Psi_{,r} = \frac{4(-r^2 + 4r - 1)}{(1+r)^4}. \quad (104)$$

The extrema satisfy $r^2 - 4r + 1 = 0$, so

$$r_{\pm} = 2 \pm \sqrt{3}. \quad (105)$$

Evaluating the kernel at those points yields

$$\Psi(r_-) = -\frac{2\sqrt{3}}{9}, \quad \Psi(r_+) = \frac{2\sqrt{3}}{9}. \quad (106)$$

These values are the ones used in Section 5 to derive the explicit baseline-frame crossing strip.

C. Memory-mode difference variable

Using Eqs. (84) and (87), the perturbative mismatch variable

$$\chi \equiv \delta\rho_X - 3c_D s \quad (107)$$

obeys

$$\chi' = \left[3c_D(1 + c_b^2) - 1 \right] \left(\nu_X + \bar{\Sigma} \frac{h'}{2} \right). \quad (108)$$

Thus the baseline closure propagates χ dynamically. The background identity $\bar{\rho}_X = 3c_D \bar{\Sigma}$ does not imply the algebraic perturbative constraint $\delta\rho_X = 3c_D \delta\Sigma$.

D. Short-wave dispersion relation

For completeness, the short-wave derivation in Section 5 is one line. In the limit $\mathcal{H} = \bar{Q} = h' = 0$, let the perturbations vary as $e^{-i\omega\eta}$. Equations (90) give

$$\nu_X = i\omega \delta\rho_X, \quad s = (1 + c_b^2) \delta\rho_X,$$

so the Euler equation becomes

$$\omega^2 \delta\rho_X = c_b^2 k^2 \delta\rho_X. \quad (109)$$

This is the hyperbolic short-wave limit used in the main text.

E. Asymptotic tail on the flat branch

When $\Psi(r) \rightarrow 0$, the sourced tail obeys

$$\dot{\rho}_X = -\frac{H}{c_D}\rho_X. \quad (110)$$

If the tail dominates the expansion, then

$$3M_{\text{Pl}}^2 H^2 = \rho_X. \quad (111)$$

Solving Eqs. (110) and (111) gives

$$a(t) \propto t^{2c_D}, \quad \rho_X(t) \propto t^{-2}. \quad (112)$$

This is the asymptotic power-law behavior quoted in Section 4.5.

F. Minimal implementation checklist

1. Add the background variables $(\Sigma, \rho_X, r, \Psi, Q)$ and, if required, the pre-confinement dark-radiation bath.
2. Verify numerically at every time step that the background identity $\rho_X = 3c_D\Sigma$ holds.
3. Implement the perturbation variables $(\delta\rho_X, \nu_X, s)$ before attempting any frame variation.
4. Confirm the analytic source-free limit and the continuous Λ CDM limit.
5. Run background-only scans to locate numerically safe regions before enabling perturbations.
6. Verify the sign change $Q < 0 \rightarrow Q > 0$ as the background passes through $r = 1$.
7. Monitor $\chi = \delta\rho_X - 3c_D s$ as part of the perturbation diagnostics.
8. Reconstruct $(\alpha_D(M_{\text{Pl}}), \xi, x_c, \varepsilon_\Sigma)$ for every posterior sample retained for physics interpretation.

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